Lab Report: Image classification experiments

1 Introduction

In this report we compare several models on standard benchmarks. The learning rate was set to 0.0003 with the Adam optimizer. All experiments were run on a single GPU with 12 GB of memory.

2 Results

Figure 1 shows the validation accuracy by dataset.

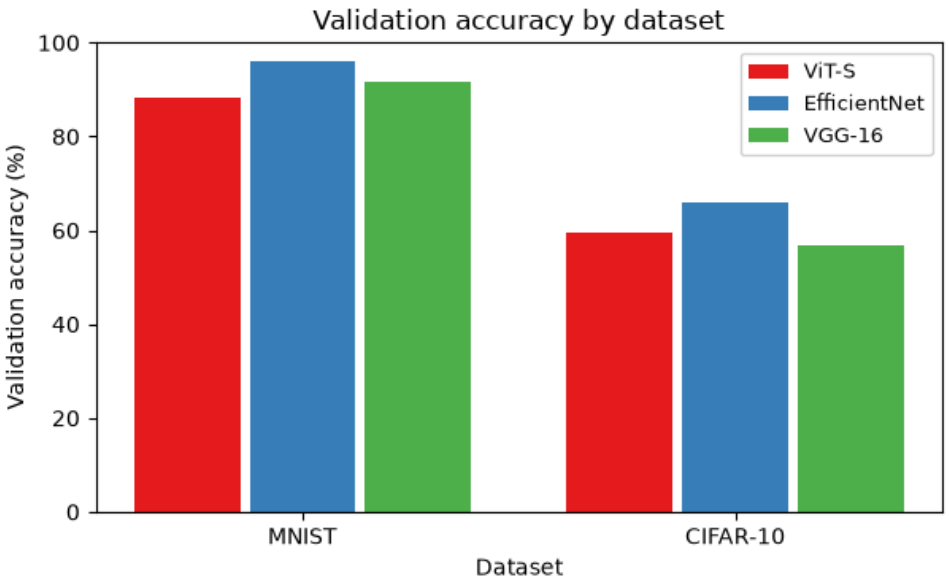


Figure 1: Validation accuracy by dataset for the models in this experiment.

EfficientNet achieves 91.7% validation accuracy on MNIST (Figure 1). On MNIST, EfficientNet is about 4 points better than VGG-16. EfficientNet outperforms ViT-S by 6.2 percentage points on CIFAR-10.

Figure 2 shows the accuracy during training.

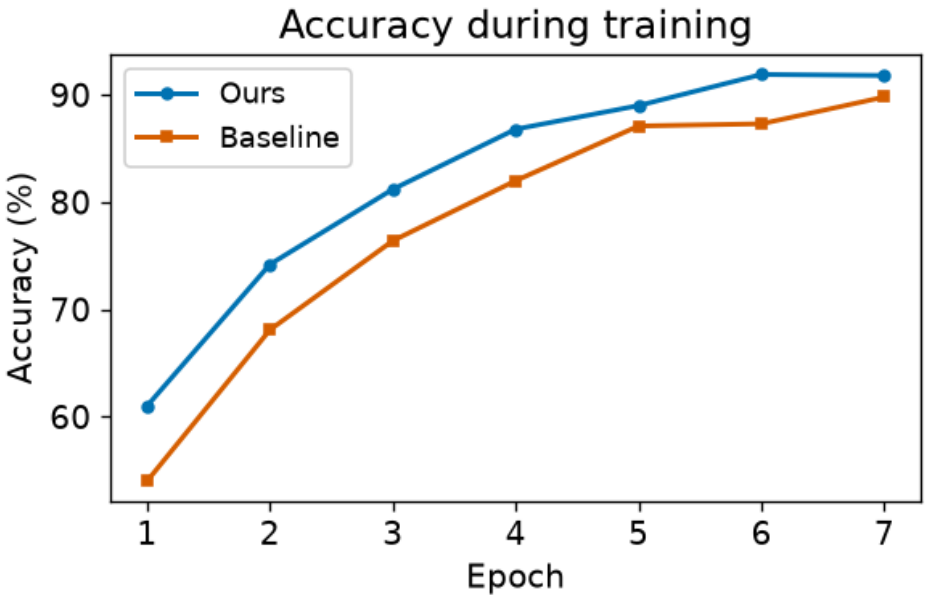


Figure 2: Accuracy during training for the models in this experiment.

At epoch 6, Ours is 3.8 points better than Baseline. From epoch 4 to epoch 7, the accuracy of Ours improves by a relative 5.8%. As shown in Figure 2, from epoch 2 to epoch 7, the accuracy of Ours improves by a relative 35.0%. As shown in Figure 2, from epoch 4 to epoch 6, the accuracy of Ours improves by a relative

6%.

Figure 3 shows the accuracy by dataset.

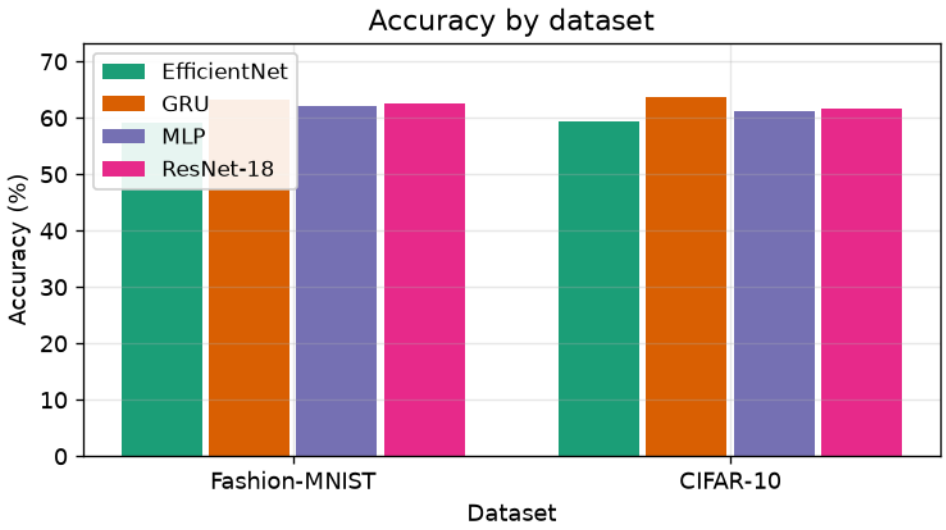


Figure 3: Accuracy by dataset for the models in this experiment.

MLP outperforms EfficientNet by 2.9 percentage points on Fashion-MNIST. On CIFAR-10, EfficientNet obtains a accuracy of 61.7%. As shown in Figure 3, MLP achieves 54.8% accuracy on Fashion-MNIST. MLP achieves over 60% accuracy on CIFAR-10.

Figure 4 shows the test accuracy of CNN by dataset.

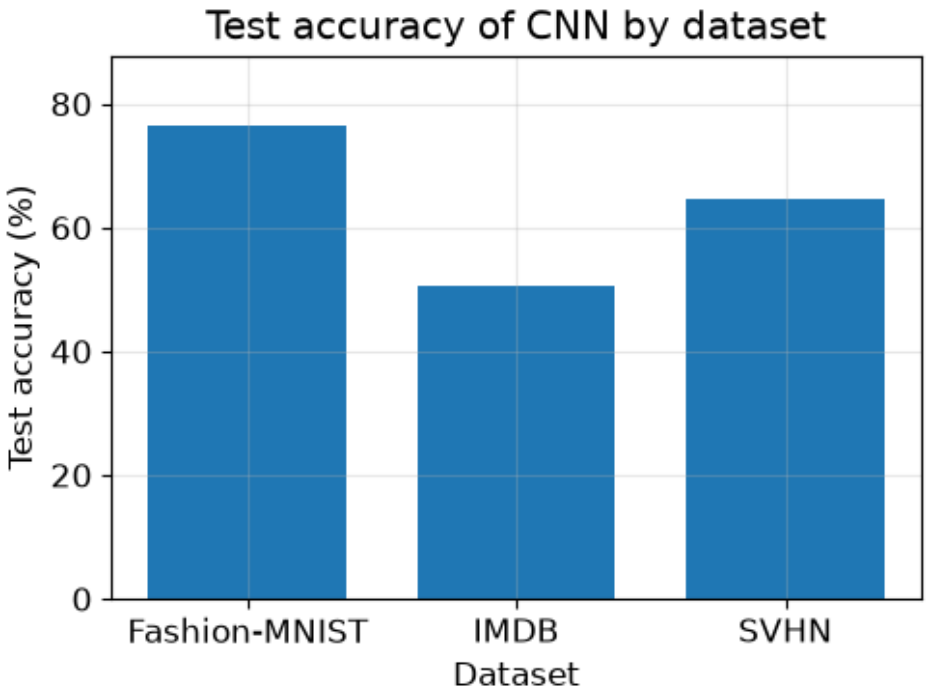


Figure 4: Test accuracy of CNN by dataset for the models in this experiment.

On IMDB, CNN obtains a test accuracy of 50.6% (Figure 4). CNN scores 43.4 points higher on Fashion-MNIST than on IMDB. CNN scores 5.3 points higher on SVHN than on IMDB (Figure 4).

3 Discussion

On SVHN, Random Forest obtains an accuracy of 90.7%. Overall, the results are in line with what we expected from the literature.